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(54) **VEHICLE DIAGNOSTIC SYSTEM AND METHOD WITH VEHICLE SYSTEM IDENTIFICATION AND CHECKLIST GENERATION**

(71) Applicant: **Opus IVS, Inc.**, Ann Arbor, MI (US)

(72) Inventors: **Brian J. Herron**, Dexter, MI (US); **Jay H. Horak**, Mesa, AZ (US); **Michael D. Jurmo**, Canton, MI (US); **David M. Joseph**, Ypsilanti, MI (US); **Michael B. Devericks**, Ventura, CA (US)

(73) Assignee: **Opus IVS, Inc.**, Dexter, MI (US)

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CPC **G07C 5/0808** (2013.01); **G07C 5/0841** (2013.01)

(58) **Field of Classification Search**
CPC G07C 5/0808; G07C 5/50841
See application file for complete search history.

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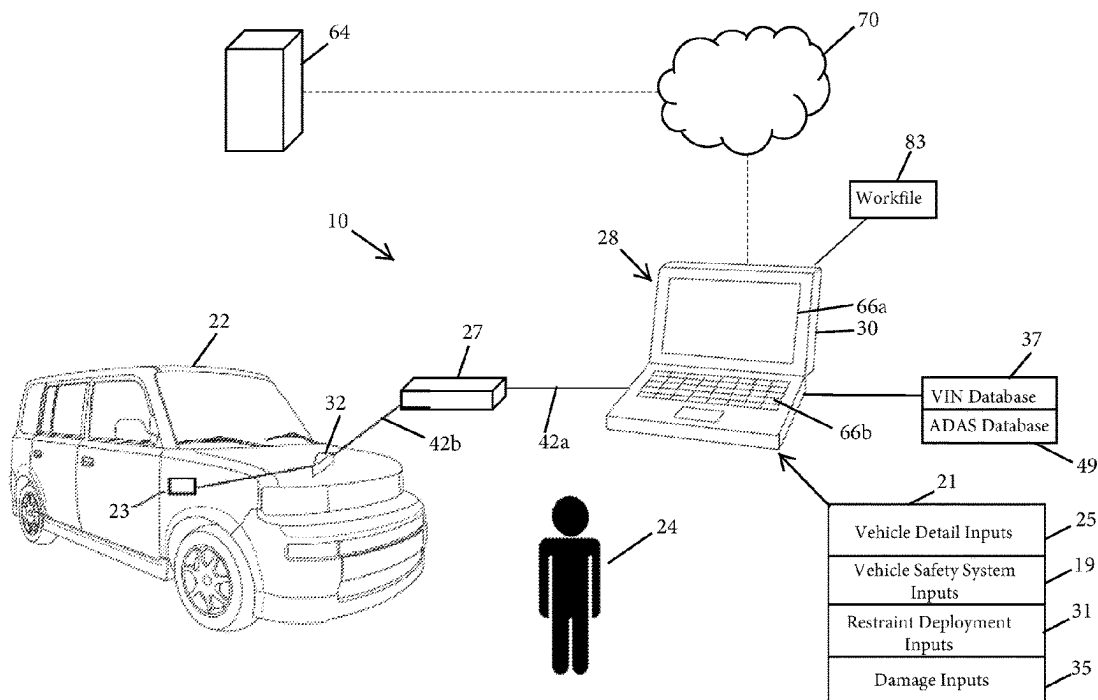
Primary Examiner — Adnan M Mirza

(74) *Attorney, Agent, or Firm* — Gardner, Linn, Burkhart & Ondersma LLP

(57) **ABSTRACT**

A vehicle diagnostic system and method for identifying and presenting vehicle safety systems on a vehicle for calibration by an operator includes providing a vehicle diagnostic computer tool that includes a display screen and is configured to be connected to a diagnostic port of a vehicle. The vehicle diagnostic tool is configured to determine vehicle safety systems present on the vehicle and to receive vehicle damage information inputs corresponding to areas on the vehicle in which there is damage. A system list program generates a list of vehicle safety systems equipped on the vehicle requiring calibration based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, where the list corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas. The vehicle diagnostic computer tool is further configured to display the list on the screen.

22 Claims, 8 Drawing Sheets



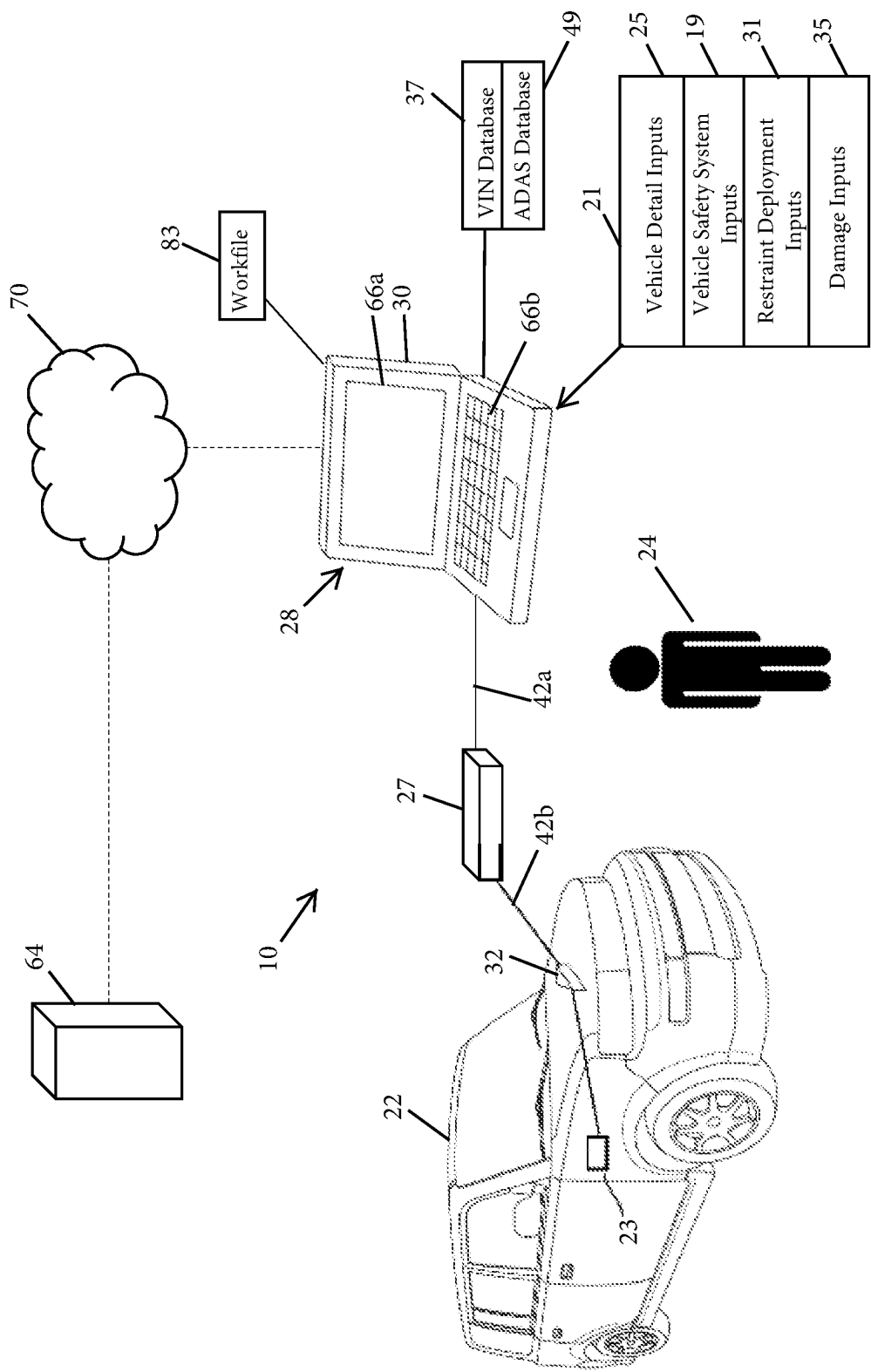
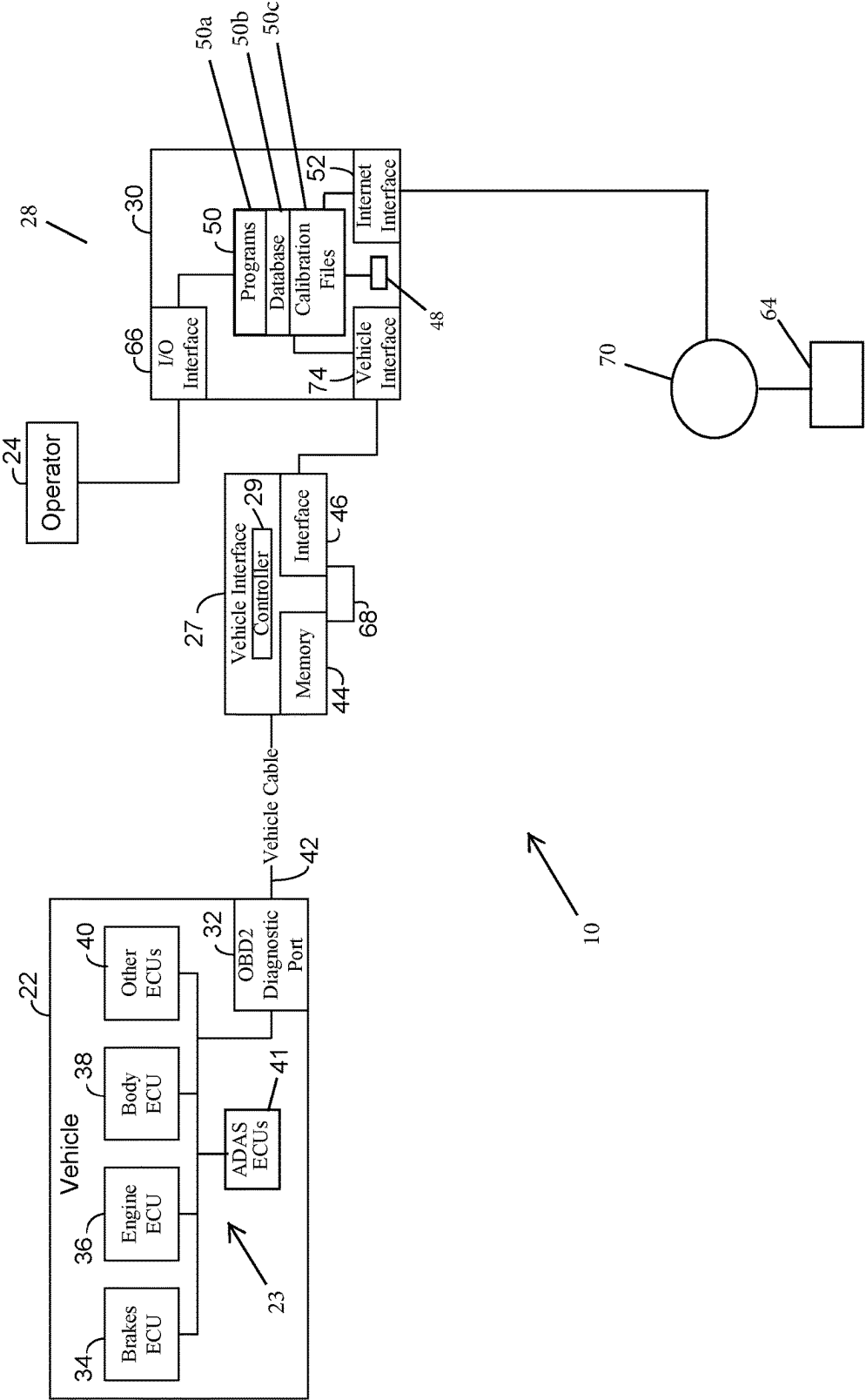


FIG. 2



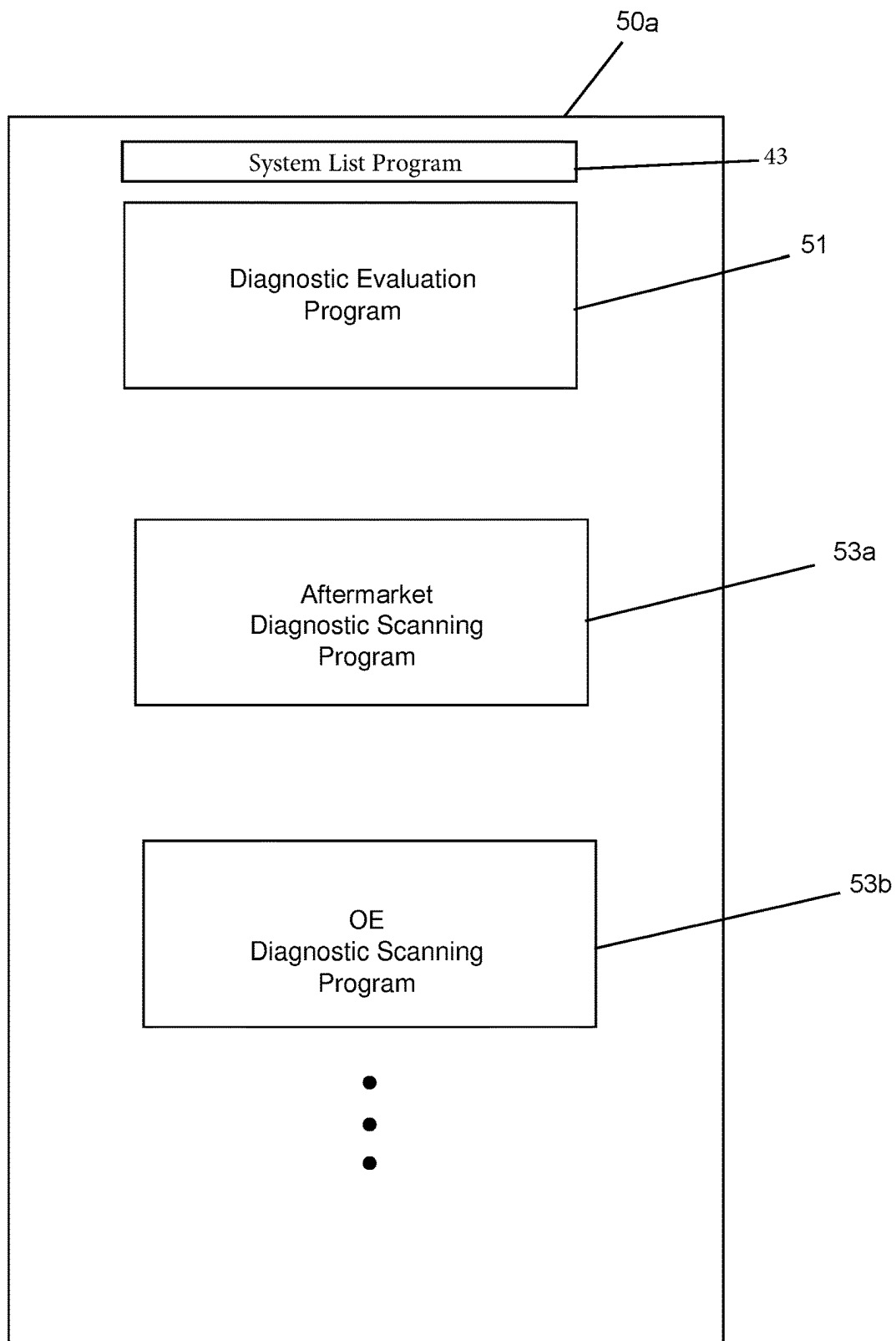


FIG. 3

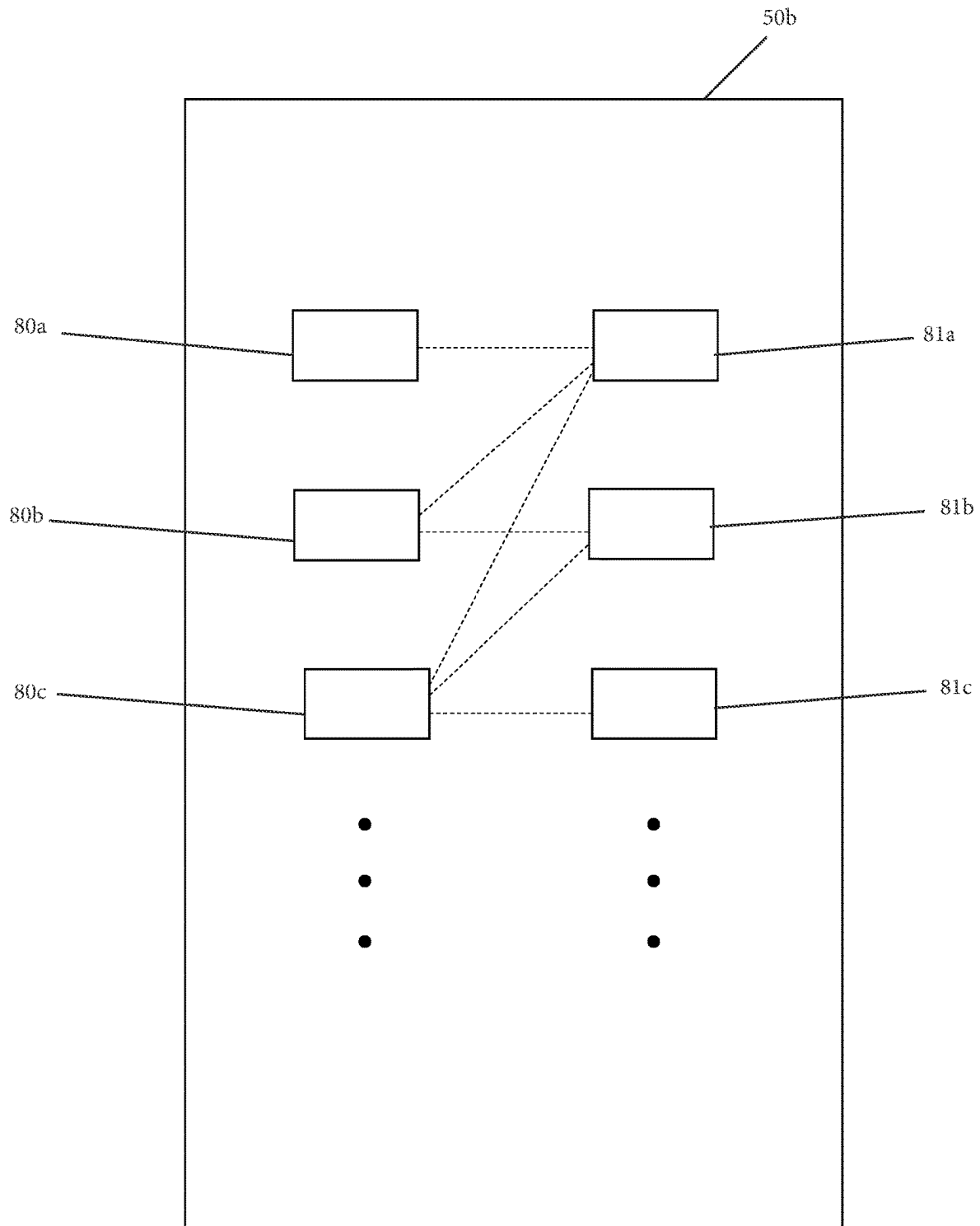


FIG. 3A

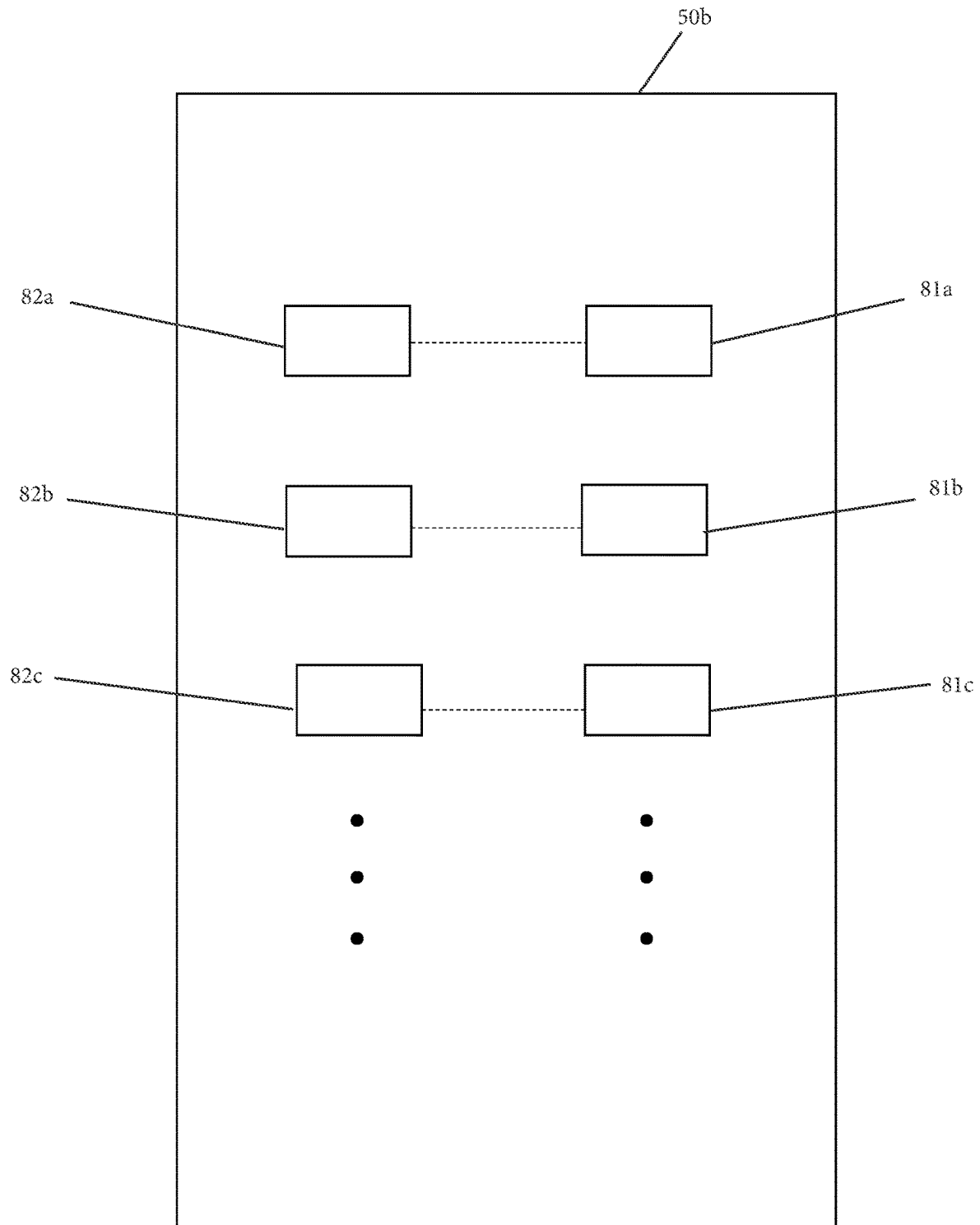


FIG. 3B

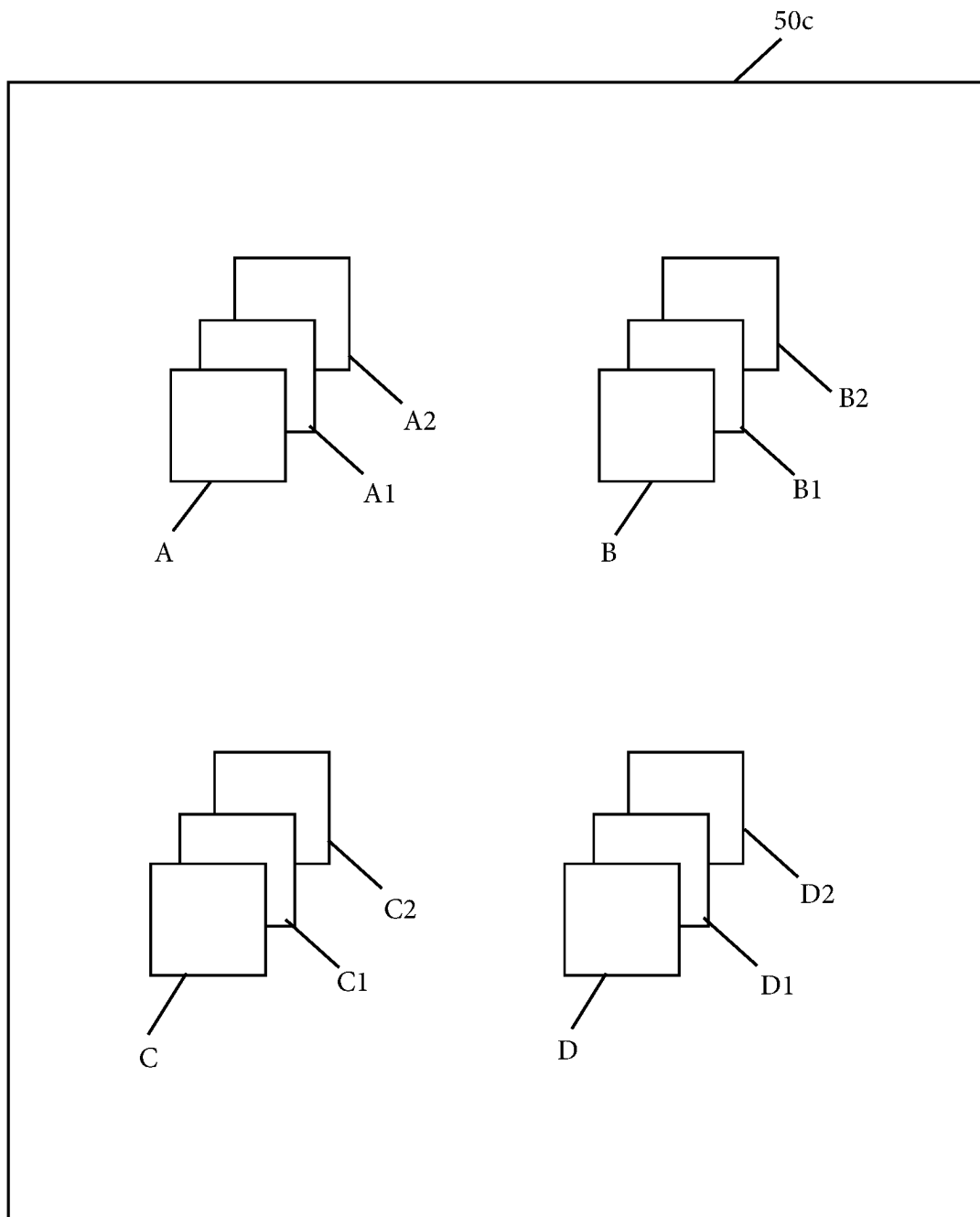


FIG. 3C

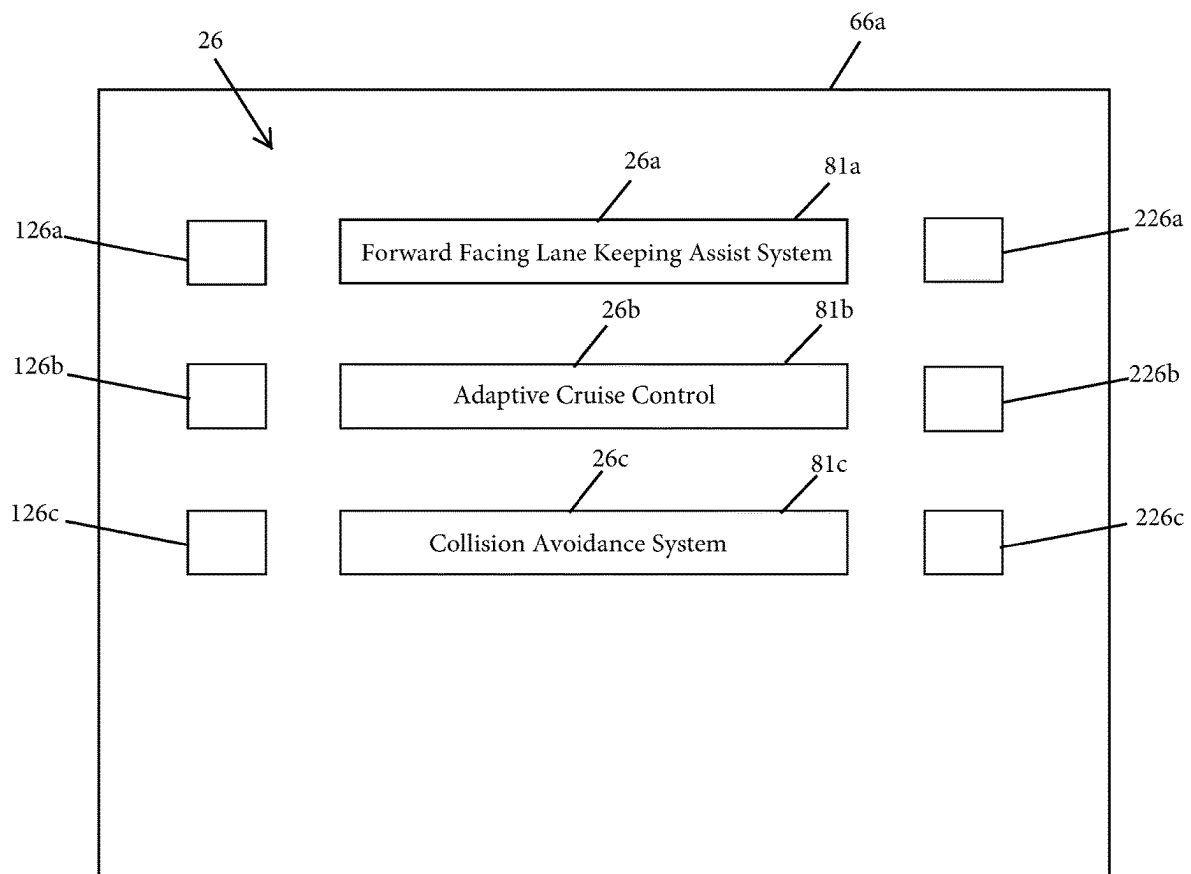


FIG. 4

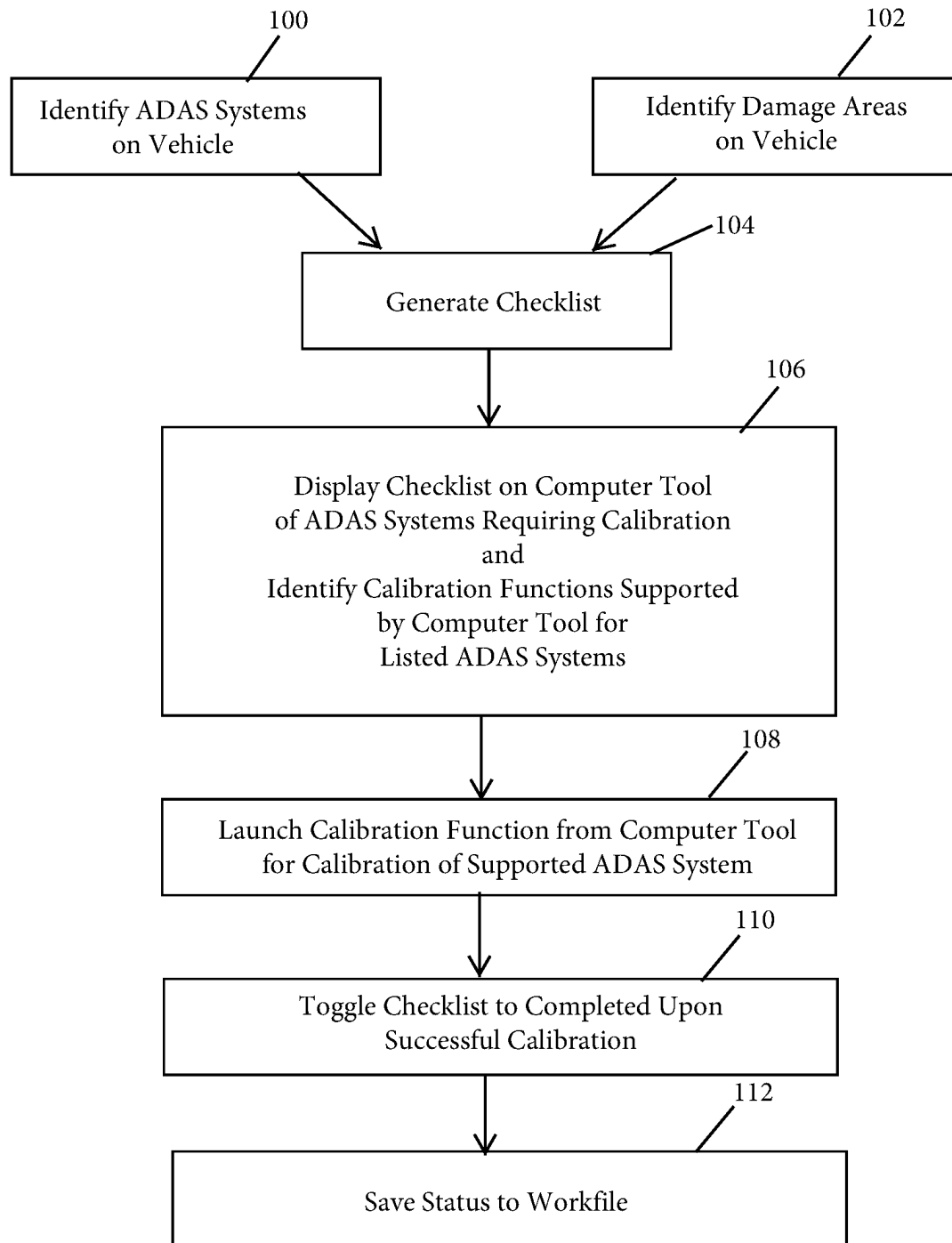


FIG. 5

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VEHICLE DIAGNOSTIC SYSTEM AND METHOD WITH VEHICLE SYSTEM IDENTIFICATION AND CHECKLIST GENERATION

CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation-in-part of U.S. application Ser. No. 17/509,316, filed Oct. 25, 2021, and is a continuation-in-part of U.S. application Ser. No. 17/515,516, filed on Oct. 31, 2021, both of which are hereby incorporated herein by reference in their entireties.

BACKGROUND OF THE INVENTION

The present invention is directed to a vehicle diagnostic system and method, and in particular to a vehicle diagnostic system that identifies electronic systems equipped on the vehicle and generates a listing of systems that may require repair or calibration based on vehicle damage and configuration, as well as identifies calibration functions supported by the vehicle diagnostic system.

Vehicle diagnostic systems employing diagnostic scan devices or tools are used in automotive repair facilities to diagnose and repair computer-based vehicle systems, where vehicles may have differing computer-based systems depending on the configuration and options installed on the vehicle. Vehicle diagnostic scan systems may include or use one or more diagnostic software scanning programs or applications, such as applications developed by an OEM or an aftermarket diagnostic company. OEM diagnostic applications may include proprietary diagnostic application software exclusive to a particular OEM, and may be required for certification purposes to validate repairs performed on a vehicle.

SUMMARY OF THE INVENTION

The present invention provides a vehicle diagnostic system and method, and in particular a diagnostic system that utilizes a vehicle diagnostic computer tool to identify electronic systems equipped on a vehicle and generate a list of systems that may require repair or calibration based on the configuration of the vehicle and any damage to the vehicle, as well as identifies calibration functions supported by the vehicle diagnostic system.

A method for identifying and presenting vehicle safety systems on a vehicle for calibration by an operator in accordance with an aspect of the present invention comprises providing a vehicle diagnostic computer tool that includes a display screen and is configured to be connected to a diagnostic port of a vehicle to be in communication with an electronic system of the vehicle, as well as includes providing a system list program that is operable to generate a list of vehicle safety systems equipped on the vehicle requiring calibration based on inputs to the system list program. The method further includes determining vehicle safety systems present on the vehicle via the vehicle diagnostic computer tool, determining areas on the vehicle in which there is damage, and providing vehicle damage information inputs to the system list program based on the damage areas. The method additionally includes generating a list of vehicle safety systems equipped on the vehicle requiring calibration via the system list program based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, where the list of

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vehicle safety systems equipped on the vehicle requiring calibration corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas, and the method further includes providing the list of vehicle safety systems equipped on the vehicle requiring calibration to a user. In one embodiment, for example, the list of vehicle safety systems equipped on the vehicle requiring calibration is provided by being displayed on a screen of the vehicle diagnostic computer tool. Alternatively, or in addition, the list of vehicle safety system equipped on the vehicle requiring calibration may be output to a report. For example, an electronic data file may be generated by the vehicle diagnostic computer tool that may be provided to another computer device and/or may be printed.

In particular embodiments, the method includes reading vehicle data information with the vehicle diagnostic computer tool and determining the vehicle safety systems present on the vehicle using the vehicle data information, where the vehicle safety systems determined to be on the vehicle may be those present on the vehicle as built or as equipped by the vehicle manufacturer. The determination of the vehicle safety systems present on the vehicle may be done via an onboard configuration determination or an offboard configuration determination, such as via one or more onboard or offboard databases. The onboard configuration determination may be obtained from the vehicle itself or via a database located on the vehicle diagnostic computer tool, and the offboard configuration determination may be performed by accessing a separate database identifying the as built configuration of the vehicle, such as by way of a vehicle identification number (VIN) of the vehicle. As such, the vehicle data information may comprise a VIN of the vehicle, and an offboard database may be used to identify vehicle systems present on the vehicle, such as by way of the VIN.

In accordance with a further aspect of the invention, the vehicle damage information inputs may comprise at least one of (i) one or more digital images of the vehicle, where image recognition software is used on the one or more digital images to determine damage to the vehicle, (ii) data from a collision estimating software, and/or (iii) data input by a user of the vehicle diagnostic tool.

Still further, the vehicle diagnostic system is configured to support calibration of one or more of the listed vehicle safety systems requiring calibration. In a particular configuration, the system displays an indication of which of the vehicle safety systems requiring calibration are supported by the vehicle diagnostic system. The vehicle diagnostic computer tool is further configured to enable a user to access and launch the calibration support information with the vehicle diagnostic computer tool for calibration of vehicle safety systems requiring calibration that are supported by the vehicle diagnostic system. The calibration support information may comprise human readable instructions contained within electronic data files stored on the vehicle diagnostic computer tool that may be displayed or printed, and may include calibration programs, such as executable programs for interacting with the electronic system of the vehicle.

The vehicle diagnostic computer tool may further be configured to require confirmation that all of the listed vehicle safety systems equipped on the vehicle requiring calibration have been calibrated prior to running a vehicle diagnostic scan with the vehicle diagnostic computer tool, such as prior to running a post scan.

In accordance with a further aspect of the present invention, a vehicle diagnostic system comprises a vehicle diagnostic computer tool that includes a display screen and is

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configured to be connected to a diagnostic port of a vehicle to be in communication with an electronic system of the vehicle, and includes a system list program. The vehicle diagnostic tool is configured to determine vehicle safety systems present on the vehicle and is configured to receive vehicle damage information inputs corresponding to areas on the vehicle in which there is damage. The system list program is configured to generate a list of vehicle safety systems equipped on the vehicle requiring calibration based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, where the list of vehicle safety systems equipped on the vehicle requiring calibration corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas. The vehicle diagnostic computer tool is further configured to display on the screen the list of vehicle safety systems equipped on the vehicle requiring calibration.

The vehicle diagnostic system and method of the present invention provides a checklist to a mechanic repairing a damaged vehicle regarding ADAS safety systems that may require calibration, as well as enables the mechanic to launch calibration functions using the vehicle diagnostic computer tool to calibrate such systems that are supported by the vehicle diagnostic system. In this way, all of the ADAS systems requiring calibration may be tracked, with calibration thereof being confirmed prior to running a post scan operation. These and other objects, advantages, purposes and features of this invention will become apparent upon review of the following specification in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram of a vehicle diagnostic system in accordance with the present invention showing a vehicle diagnostic computer tool connected to the electronic system of a vehicle via a vehicle interface device;

FIG. 2 is a block diagram of the vehicle diagnostic system in accordance with the present invention in relation to the electronic system of the vehicle;

FIG. 3 is a block diagram of programs stored in memory, including a system checklist program, diagnostic evaluation program, and diagnostic scanning program applications of the vehicle diagnostic system of FIG. 1;

FIG. 3A discloses an exemplary database correlating vehicle identification number data to vehicle systems for a vehicle;

FIG. 3B discloses an exemplary database correlating vehicle identification number data to vehicle systems for a vehicle;

FIG. 3C discloses exemplary calibration files for calibration of vehicle ADAS systems;

FIG. 4 illustrates a vehicle system checklist displayed to a mechanic on a computer device that is generated by the vehicle diagnostic system in accordance with aspects of the present invention; and

FIG. 5 is a flow chart illustrating aspects of the vehicle diagnostic method in accordance with the present invention by which the vehicle diagnostic computer tool provides a system checklist to an operator.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention will now be described with reference to the accompanying figures, wherein the numbered

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elements in the following written description correspond to like-numbered elements in the figures.

A vehicle diagnostic system 10 for use with a vehicle 22 is shown for use by a user such as a mechanic or operator 24, such as in an automotive repair facility. System 10 is illustrated in FIGS. 1 and 2 as including a vehicle diagnostic computer tool 28, which as discussed in more detail below, is connected with vehicle 22 by operator 24 in order to diagnose the electronic system 23 of vehicle 22, including various vehicle electronic control units (ECUs), including safety systems such as Adaptive Driver Assistance Systems ("ADAS") ECUs, and including other electronic parts and components of vehicle 22. Vehicle diagnostic tool 28 may be used to scan the electronic system 23 of vehicle 22, such as to determine any fault codes in the electronic system 23, which may be reported as diagnostic trouble codes ("DTCs"). A vehicle scan may be run prior to repairs, such as part of an initial diagnosis to provide an indication to the mechanic 24 as to what repairs are needed on vehicle 22, and/or may be run after performing repairs, such as to confirm that vehicle 22 has been properly repaired.

As discussed in more detail below, diagnostic computer tool 28 is configured to determine the electronic vehicle systems with ECUs equipped on vehicle 22, such as in particular safety systems, including ADAS systems, and is further configured to receive inputs 35 corresponding to damage to vehicle 22, such as from a collision, with the diagnostic computer tool 28 in turn generating a vehicle system list or system checklist 26 (FIG. 4) identifying and presenting to the mechanic 24 electronic vehicle systems of vehicle 22 that may need calibration, where the status of completion of the indicated calibrations can be readily tracked and recorded to confirm completion of repairs. According to still additional features, diagnostic computer tool 28 may support calibration functions of various electronic vehicle systems of vehicle 22, with the system checklist 26 in turn including an identification for the mechanic 24 of the supported calibration functions from which the mechanic 24 can select and launch using diagnostic computer tool 28.

As discussed in more detail below, diagnostic system 10 utilizing diagnostic computer tool 28 is thus operable based on inputs to diagnostic tool 28 to provide a checklist 26 to mechanic 24 of systems on vehicle 22 with ECUs, such as safety systems including ADAS systems and airbag modules, that may require calibration based on the determined configuration of vehicle 22 and damage thereto, with diagnostic tool 28 additionally identifying for mechanic 24 any such systems for which diagnostic tool 28 supports calibration for that vehicle 22. The checklist 26 is generated by a system list or system checklist program 45 that may additionally be configured to communicate with one or more remote computer systems, such as with a sever 64 via an Internet connection 70, including for example to confirm completion of calibrations and to automatically toggle items on the checklist 26 as completed when calibrations have been performed. It should be appreciated, however, that diagnostic computer tool 28 may itself be able to confirm completion of calibrations and automatically toggle items on checklist 26, such as for calibration operations supported on diagnostic computer tool 28.

Vehicle diagnostic computer tool 28 is configured in the illustrated embodiment as comprising a computer device 30, such as a laptop or tablet computer that includes circuitry, hardware and software, and is coupled with a vehicle interface module 27 by a cable 42a. In use, tool 28 is connected with vehicle 22 by operator 24, such as by

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connecting to an on-board diagnostic (“OBD”) diagnostic port 32 of the vehicle 22 in order to diagnose the electronic system 23 of vehicle 22, including the various noted vehicle systems comprising vehicle electronic control units (ECUs), such as an engine ECU 34, body ECU 36, brakes ECU 38, and/or other ECUs 40, including for safety systems such as ADAS ECUs 41, and including other electronic parts and components of vehicle 22, including other safety systems such as airbag modules and the like. Tool 28 connects with port 32 via interface 27, such as via vehicle cable 42b. Vehicle interface 27 includes a controller 29, such as in the form of a processor or micro-processor and interface circuitry to facilitate communication between the ECUs and tool 28, with interface 27 including a database of vehicle protocols found in a local memory 44 that allow communication with the ECUs of various makes and models of vehicles. Vehicle interface 27 additionally includes an interface 46 for communication between interface module 27 and computer 30 via interface 74 of computer 30, where computer 30 additionally includes a controller 48 and memory 50. Diagnostic tool 28 additionally includes a user interface 66, which may comprise a touch screen 66a and/or keyboard 66b of computer 30 enabling mechanic 24 to enter information into diagnostic tool 28, as well as view information output by diagnostic tool 28. As understood from FIGS. 1 and 2, vehicle diagnostic tool 28 may be connected to a remote computer, such as a server 64, such as by an Internet 70 connection. Although vehicle diagnostic computer tool 28 is shown as comprising a separate computer 30 coupled with vehicle interface device 27 in the illustrated embodiment, it should be appreciated that in an alternative arrangement computer 30 and vehicle interface device 27 may be integrated together, such as for example, in an arrangement such as disclosed in commonly assigned U.S. patent application Ser. No. 16/701,967, which is incorporated herein by reference in its entirety.

Vehicle diagnostic system 10 may be configured to be selectively operable in one of a plurality of different modes, whereby a technician may use the diagnostic tool 28 for vehicle maintenance, diagnosis, programming and repair as needed. In the illustrated embodiment, and as understood from FIGS. 2 and 3 in accordance with an embodiment of the present invention, system 10 includes or provides access to various programs 50a comprising diagnostic applications, such as stored in a memory 50 of diagnostic computer tool 28, including system list program 45, a diagnostic evaluation program application 51, and commercially available diagnostic application scanning programs 53a, 53b that are configured for use with the specific vehicle 22 and may be used or selected by a mechanic 24. In the illustrated embodiment diagnostic application program 53a comprises an aftermarket scan program and diagnostic application program 53b comprises an OE scan program. Although shown as having two diagnostic application programs 53a, 53b, it should be appreciated that system 10 may include more than two such programs available for use with a given vehicle 22 and/or including additional diagnostic application programs for use with other vehicles, depending on make, model and/or year of a given vehicle. Interface module 27 may be configured as an SAE standard J2534 device, such as a device compliant with the J2534-2 standard, or as an ISO compliant or other standard compliant device for supporting and enabling communication with the electronic systems of a vehicle.

As illustrated in FIGS. 2 and 3, the diagnostic application scanning programs 53a, 53b are stored in a memory 50. An exemplary aftermarket diagnostic application scanning pro-

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gram 53a comprises a program provided by an automotive manufacturer or a company that supplies diagnostic application programs, such as Opus IVS, Inc. of Ann Arbor, Michigan, and the OE diagnostic application scanning program 53b comprises a program provided by a domestic or foreign OEM, such as Ford, General Motors, Toyota or the like, with the diagnostic application scanning tool 28 configured to enable the reading and reporting of fault codes in the electronic system of the vehicle such as may be located in ECUs of the vehicle. As noted, in practice, memory 50 may include multiple diagnostic application programs, each for use with various makes and/or models of vehicles to enable diagnosing and programming of ECUs via vehicle interface module 27, including depending on the particular vehicle systems/ECUs on the vehicle installed by the OEM based on the vehicle purchaser's selection of vehicle options. Alternatively and/or additionally, memory 50 may include diagnostic application programs that may be used with multiple variations of vehicles. Vehicle interface module 27 and computer 30 are thus cooperatively used for querying/scanning and diagnosing ECUs of vehicle 22, including for accessing error codes generated by the ECUs for assessing and diagnosing operational and performance related aspects of the vehicle 22.

As illustrated in FIG. 1, diagnostic tool 28 may receive various inputs 21 that are used, including by system list program 45, to generate the checklist 26. In the illustrated embodiment this may include vehicle detail inputs 25, vehicle safety system configuration inputs 19, restraint deployment status inputs 31, and vehicle damage inputs 35.

Vehicle detail inputs 25 may comprise various information, including the make, model and/or year of the vehicle 22. The vehicle detail inputs 25 may further or alternatively include the vehicle identification number (“VIN”) of the vehicle 22. The vehicle detail inputs 25 may be manually entered by mechanic into diagnostic tool 28, such as via touch screen 66a or keyboard 66b. Alternatively, vehicle detail inputs 25 may be obtained by diagnostic tool 28 by reading from electronic system 23, where diagnostic tool 28 may query electronic system 23, such as to obtain the VIN. System 10 may further employ a VIN database 37 via which specifics of vehicle 22 may be ascertained by diagnostic tool 28. For example, VIN database 37 may be used to determine the make, model and/or year of vehicle 22 via the VIN.

Safety system inputs 19 are provided to or are determined by diagnostic computer tool 28 to identify the safety systems equipped on the vehicle, including ADAS systems and airbag modules on vehicle 22, such as the safety systems on the vehicle as built or equipped by the vehicle manufacturer when built, or the existing safety systems on the vehicle. It should be appreciated that different vehicles may be equipped with different ECUs when built, including that some vehicles of the same make/model may have different or additional ECUs when compared to other vehicles of the same make/model, such as based on options or equipment packages. For example, a given vehicle may be sold with or without various advanced driver-assistance systems (ADAS), such as adaptive cruise control, lane departure warning, parking assistance, blind spot detection, collision avoidance, forward collision warning, surround view, automatic parking, and other driver assistance systems or other vehicle options. Each ADAS subsystem may include its own additional ECUs, or an ECU may be shared for more than one ADAS feature, where such additional ECUs must be queried/scanned, accessed, and then assessed. In particular, such safety systems may require calibration, such as to confirm proper operation, including if such systems required

repair or replacement as a result of collision damage. In accordance with aspects of the present invention, therefore, prior to generating a checklist 26 for providing a mechanic 24, an evaluation may be performed to determine the vehicle systems present on the subject vehicle whereby the checklist 26 may be generated to only include systems that are on the specific vehicle 22 at issue, including based on the particular systems/ECUs present on the vehicle and/or any particular damage that occurred to vehicle 22, such as from a collision. The configuration of vehicle safety systems of a vehicle may be ascertained via an onboard or offboard determination.

Safety system inputs 19 may be obtained in a number of ways, including using either one or more onboard databases or offboard databases. In one onboard configuration, diagnostic tool 28 includes a diagnostic evaluation program application 51 (FIG. 3) for querying/scanning the vehicle 22 to read vehicle data information regarding the particular ECUs that are present on the vehicle, with the vehicle data information being analyzed or parsed to determine the particular vehicle systems provided on the vehicle based on the particular ECUs that are detected. In particular, system 10 determines whether vehicle 22 includes particular safety systems, such as any ADAS systems based on the particular ECUs that are detected. Accordingly, prior to generating a checklist 26 for a given vehicle, a diagnostic evaluation program 51 may initially be run to read or obtain vehicle data information related to the equipped vehicle systems on the vehicle. In one embodiment, vehicle diagnostic tool 28 includes an onboard memory or database 50b that includes known ECUs that may be present on a vehicle, including ECUs for ADAS systems on a vehicle. Diagnostic evaluation program 51 is run to determine the ECUs that are present on the vehicle 22, where the detected ECUs may be compared to the known ECUs identified in database 50b that may be present, whereby the particular ECUs present on vehicle 22 may be identified. In one embodiment, information contained within database 50b includes information regarding addresses within electrical system 23 for the potential ECUs of vehicle 22, where the addresses may be specified, for example, by a bus address, such as a unique memory address or CAN address for the various addresses of the ECUs. For example, database 50b may include information regarding each possible ECU that was available for a given make and model of vehicle, such as by year, including for all potential options, including ADAS systems, including the specific ECU for each possible module and the address within electrical system 23 that such ECU is located. Diagnostic evaluation program 51 may operate to query each address of electrical system 23 at which an ECU may be present, such as by sending an inquiry signal to such addresses, where system 10 is able to confirm the presence of a given ECU for such addresses if a response signal is received from an ECU at the given address. Moreover, system 10 is able to identify the particular ECU, including whether the ECU is an ADAS ECU based on the address, which identification information may be stored in database 50b along with the given address. For example, each potential ECU of a given make and model of vehicle, including by year or years, may be mapped out such that database 50b includes the address of each ECU, and may also include the specific operation, purpose or function of the ECU, such as by part number, name, or the like. System 10 may serially query addresses of electrical system 23 to determine the presence of particular ECUs, including ADAS ECUs, or may alternatively broadcast inquiry signals to multiple or all potential addresses in parallel. Still further, rather than query each ECU within the electronic system 23, system 10 may

be configured to query only for ADAS ECUs. In such a configuration database 50b may only contain ECU addresses for ADAS systems. Still further, as discussed in more detail below, a remote or offboard database may be employed for determining the safety systems present on vehicle 22, where for example, a database such as database 50b may be disposed at a remote computer, such as server 64.

In an alternative configuration or operation, system 10 may obtain safety system inputs 19 by determining the ADAS systems present on vehicle 22 by way of build data from the vehicle database 50b that comprises a VIN database. For example, the diagnostic evaluation program 51 may obtain the vehicle VIN upon connection of system 10 with vehicle 22. As a first step, system 10 via the vehicle diagnostic tool 28 may initially acquire a vehicle identification number (VIN) associated with a vehicle under test. In an aspect of the present invention, the diagnostic tool 28 is operable to read the VIN from the vehicle 22 via its connection through the OBD2 diagnostic port 32. The vehicle VIN may then be used to determine the vehicle systems equipped on the vehicle, such as via VIN database 50b, where for example, computer 30 may include database 50b that is operatively used to determine the vehicle systems on the vehicle based on the determined VIN. This may include an algorithmic lookup table based on the identified VIN, such as for example where certain alphanumeric characters of the VIN identify the presence or absence of particular vehicle systems present on the vehicle, such as ADAS systems or other vehicle or safety systems. Alternatively, the VIN for the vehicle 22 may be acquired through alternative means and directly input by the operator 24, such as by being visually examined and input via interface 66, such as by way of a keyboard or touch screen. Moreover, rather than a lookup VIN database, the system 20 may operatively algorithmically analyze selected alphanumeric characters, such as by position number in the VIN, with the system 20 recognizing based on the particular character and location the presence or absence of particular vehicle systems present on the vehicle, such as being preprogrammed. This may include, for example, an operator initially entering a make and model of a vehicle via interface 66 whereby the system 20 is preconfigured to read particular characters in particular locations of the VIN in order to determine the presence or absence of particular vehicle systems present on the vehicle. Again, as noted below, rather than being configured as an onboard database, database 50b may alternatively comprise an offboard database, such as residing at remote server 64.

An exemplary embodiment of a database 50b configured as a VIN database is illustrated in FIG. 3A, such as for one particular make and model of vehicle. As there shown, database 50b includes a listing of VIN data 80a, 80b, 80c, where VIN data may comprise complete VIN numbers, ranges of VIN numbers, or selected portions or ranges of VIN numbers for a given make and model vehicle, with the VIN data 80a, 80b, 80c being correlated with vehicle systems 81a, 81b, 81c, where the vehicle systems 81a, 81b, 81c in the illustrated embodiment are ADAS systems such as a forward facing lane keeping assist system, an adaptive cruise control system, and a collision avoidance system. It should be appreciated that although database 50b is illustrated in FIG. 3A to include three separate VIN data designations and three separate vehicle systems, that in practice the VIN database may include numerous listings of VIN data as well as numerous vehicle systems. The vehicle database 50b may further comprise a VIN/ECU part number database that includes a listing of ECU part numbers associated with

a given VIN, such as based on build data, including such as may be provide by an OEM. Accordingly, generation of checklist **26** may be based in part upon a review of the VIN database and/or a review of the ECU part numbers associated with the current vehicle's VIN.

Still further, safety system inputs **19** comprising the ADAS systems present on a vehicle may be obtained by way of vehicle database **50b** based on the year, make and model of a vehicle **22**. In such a configuration the vehicle database **50b** may identify the possible ADAS systems on vehicle **22** based on the year, make and model of vehicle **22**. As noted above, this may be acquired via diagnostic tool **28** reading the electronic system **23**, or may be entered by the mechanic **24**, such as via screen **66a**.

Although database **50b** is shown as being within memory **50** of computer device **30**, it should be appreciated that database **50b** may alternatively be remotely located, such as at server **64** and accessible via internet connection **70**.

It should be further appreciated that in the step of determining the presence of the particular ECUs on the vehicle **22**, that the absence of a response signal from an ECU at a given address, such as an ADAS ECU, may mean either that the module and ECU was not present on the vehicle, i.e. it was not originally equipped with the ADAS module and associated ECUs, or that the module having such ECU has been damaged. For example, in the case of a vehicle that has been damaged, such as by a collision, it is possible that the vehicle may have been equipped with particular vehicle systems or modules having ECUs, such as ADAS ECUs, but that due to the damage to the vehicle the diagnostic evaluation program **51** may not be able to detect all of the ECUs due to the vehicle damage. For example, a vehicle may be equipped with certain exterior cameras, radar or ultrasonic sensors, or other ADAS equipment. If such components are damaged in a collision, and or associated controllers are damaged in a collision, then diagnostic evaluation program **51** may inadvertently infer that such systems were not present on vehicle when instead they are not responding or detectable due to the damage. As such, in accordance with a further aspect of the present invention, inputs **21** of system **10** may additionally include inputs **35** related to damaged areas of vehicle **22** whereby system **10** takes into consideration the damage and the potentially effected vehicle systems and associated ECUs, including ADAS systems and associated ADAS ECUs. With reference to FIG. 1, damage information inputs are noted at **35** in FIG. 1.

Damage information inputs **35** may be provided to system **10** via one or more various sources. In one configuration, one or more images representative of the particular vehicle **22** being diagnosed may be shown on screen **66a** of diagnostic tool **28** where, for example, based on a detected VIN, diagnostic tool **28** may display an image representative of the make, model and year of vehicle **22**. The mechanic **24** may then be prompted to interact with the displayed image, such as with screen **66a** being configured as a touchscreen, to designate or highlight on the displayed image the location or locations corresponding to any actual damage on vehicle **22**. Alternatively, in another configuration system **10** may obtain digital images or photographs of the actual vehicle **22** requiring repair and, via image recognition software, determine the location or locations of damage on vehicle **22**. Such digital images may be provided to system **10** from a separate camera, or for example, diagnostic tool **28** may include an integrated camera or imager, such as a CMOS imager, with which to take digital images of vehicle **22**, with image recognition software residing in computer module **30**, such as in memory **50**. Still further, in yet another configuration

system **10** may interface with a collision estimating software program, such as via an API exchange, whereby system **10** may receive a predetermined evaluation of any damage to vehicle **22**, including to specific vehicle systems whereby the damaged ADAS modules are determined. For example, information regarding physical damage to a vehicle via is illustrated as being provided to diagnostic tool **28** from a collision estimating software residing on a separate computer **65** in FIG. 1. It should be appreciated that the collision estimating software may alternatively reside on diagnostic tool **28** itself, or be accessed directly via diagnostic tool **28**.

As also understood from FIG. 1, an additional input **21** comprises restraint deployment inputs **31**. Restraint deployment inputs **31** include an identification of whether or not one or more of the vehicle airbags have been deployed, which may be entered by a mechanic **24** into diagnostic tool **28** by way of a visual inspection of vehicle **22**, with diagnostic tool **28** providing a prompt to direct and allow entry of a response by mechanic **24**. Alternatively, diagnostic tool **28** may be used to perform or run a pre-scan diagnostic application program, which may be a limited pre-scan, that determines if any trouble codes for the vehicle airbags are detected indicating that airbags on vehicle **22** have been deployed. The restraint deployment inputs **31** may additionally include an identification of whether or not one or more of a seat belt tensioner or pretensioner has been activated or deployed. For example, some vehicles include electronic seat belt pretensioners that utilize an explosive charge and include an ECU for providing tension to a seatbelt in the event a collision is detected. Accordingly, in like manner to determining deployment of airbags, deployment of seatbelt pretensioner restraints may be determined by visual inspection by mechanic **24** and entry into diagnostic tool **28**, or by use of diagnostic tool **28** to perform a pre-scan to detect trouble codes indicating deployment of the seatbelt pretensioner system.

It should be appreciated that the ADAS system identification herein for purposes of the present invention may include or encompass identification of airbag modules requiring programming, and that calibration for purposes of the present invention may include or encompass the programming of the airbag modules.

Upon determining the safety systems and damage present on vehicle **22**, system list program **45** is operable to generate checklist **26** that in one embodiment is presented to mechanic **24** on screen **66** of computer **30**. As understood from FIG. 4, in the illustrated embodiment checklist **26** includes buttons or display fields **26a**, **26b**, **26c** listing various vehicle systems **81a**, **81b**, **81c**, which comprise ADAS safety systems that have been identified via system list program **45** as being present on vehicle and associated with damage to vehicle **22**, and which accordingly may require calibration operations to be performed by the mechanic **24**, where the systems **81a**, **81b**, **81c** would include various ADAS ECUs **41**. That is, the vehicle systems **81a**, **81b**, **81c** are or include components that are disposed in locations in which damage has been identified on vehicle **22**, where in the illustrated embodiment the ADAS systems comprise a forward facing lane keeping assist system **81a**, an adaptive cruise control system **81b**, and a collision avoidance system **81c**. It should be appreciated that each of these ADAS systems may comprise numerous components, including for example various cameras, radar modules and associated ECUs, and that various of such components may be utilized on more than one given ADAS system. Calibration operations for the ADAS systems **81a**, **81b**, **81c** may be performed by the mechanic **24** such as part of the repair

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and/or replacement of components of such ADAS systems and/or to confirm operation of the systems.

Checklist 26 in the illustrated embodiment additionally includes buttons or display fields 126a, 126b, 126c associated with each of the ADAS systems 81a, 81b, 81c that are operable to provide an indicator to the mechanic as to whether or not each of the systems 81a, 81b, 81c have been calibrated, where fields 126a, 126b, 126c may be toggled from not calibrated to completed upon performance of the appropriate calibration or calibration procedure for the given ADAS system. Still further, checklist 26 may include additional buttons or fields 226a, 226b, 226c associated with each of the ADAS systems 81a, 81b, 81c displayed on the checklist 26 that indicate whether or not the diagnostic computer tool 28 supports calibration of the associated ADAS system 81a, 81b, 81c. Moreover, if diagnostic computer tool 28 supports calibration of the given ADAS system, mechanic 24 may click the associated button 226a, 226b, 226c for that supported ADAS system and launch directly to the calibration function or support function for the given ADAS system. This beneficially avoids the mechanic 24 from having to navigate through to alternative programs, including avoiding having to enter information to access the appropriate calibration materials for a given ADAS system.

For example, with reference to FIGS. 2 and 3C, diagnostic computer tool 28 may include stored electronic calibration data 50c retained within memory 50 for support of calibration of one or more ADAS systems, such as systems 81a, 81b, 81c. It should be appreciated that the calibration operation for a given ADAS system 81a, 81b, 81c may be dependent upon the given system as well as the year, make and model of vehicle 22 at issue. Accordingly, memory 50 may include distinct electronically stored calibration data or information, such as shown at A, B, C, D in FIG. 3C, for use in calibrating various different ADAS systems, where the stored calibration data may comprise one or more various calibration related instruction files or programs, such as illustrated as A1, A2, B1, B2, C1, C2, D1 and D2. Although shown in the illustrated embodiment as being located in memory 50, it should be appreciated that some or all of the calibration data 50c may be remotely located, such as in server 64, and accessible via Internet connection 70.

In operation, the electronic calibration data 50c provides instruction files to a mechanic 24 comprising human readable information that may be displayed on screen 66a or printed for calibration of the given ADAS system and/or provides executable calibration programs that interact with ECUs of the ADAS system for conducting, completing and confirming calibration of the ADAS system. With respect to the illustrated embodiment of FIG. 4, for example, fields 226a, 226b, 226c indicate that diagnostic computer tool 28 supports calibration operations for the given ADAS systems 81a, 81b, 81c indicated at 26a, 26b, 26c as requiring calibration. Mechanic 24 may launch particular calibration support data 50c by selecting one of the associated fields 226a, 226b, 226c, where this may be done in any of various manners, such as via depressing a touchscreen or selecting via a cursor such as with a mouse. Upon selecting one of the associated supported fields 226a, 226b, 226c to directly launch an associated calibration instruction file, screen 66a may provide instructions to mechanic 24 regarding the calibration procedure for the given ADAS system. This may include, for example, providing a list of step-by-step instructions for the mechanic 24 to undertake to calibrate the given ADAS system, this includes identifying the particular equipment needed, such as targets for sensors of the ADAS

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system, as well as the location in which to position the targets and/or vehicle relative to each other. Still further, electronic calibration data 50c may include calibration programs, such as OEM and/or aftermarket ADAS calibration programs, for running on ADAS ECUs 41 as part of the calibration process. For example, the ADAS calibration program may be run via vehicle interface 27 on electronic system 23. Diagnostic computer tool 28 may also require or prompt mechanic 24 to verify completion of each given step, such as by way of entry via screen 66a and/or keyboard 66b. Completion of the various steps may be confirmed in various ways, including for example where completion of some or all of the steps is automatically monitored by diagnostic computer tool 28 and/or by confirmation entries by mechanic 24 via diagnostic computer tool 28. The ADAS calibration routine may additionally include running of calibration programs, such as program A1 for a given ADAS system. The calibration program may interface with an ECU in any of various known manners, including sending a command to begin a learn function with diagnostic computer tool 28 waiting for a response from the ECU indicating completion, or the calibration program may transmit persistent messages to an ECU for calibration, or the calibration program in diagnostic computer tool 28 may trigger or activate an internal OEM program of the ECU used for calibration.

Upon completing all of the steps and running of the associated calibration programs, diagnostic computer tool 28 may be returned to display checklist 26, which is then updated to reflect that a successful calibration of the ADAS system has been performed. For example, the mechanic 24 may navigate to the checklist 26 screen via computer 30 and manually toggle the associated field 126a, 126b or 126c of checklist 26 to indicate that the calibration has been performed. Alternatively, upon successfully calibrating the given ADAS system, system 10 may automatically return to the screen 66a display of the checklist 26 and automatically toggle the associated field 126a, 126b, or 126c upon completion of the calibration event. System 10 may then additionally record the successful calibration event, or indication thereof, to a work file 83 (FIG. 1), where work file 83 may be retained in memory on computer 30 and/or server 64.

System 10, such as via diagnostic computer tool 28, may additionally transmit a report comprising work file 83 or based on work file 83, such as to remote computer device 64 or another remote computer. For example, such a report may be transmitted to a shop owner, insurance estimator, insurance company, the vehicle owner, or the like. The report may include indication of successful completion of the calibration of all ADAS systems identified in the checklist 26 as requiring calibration. The report may additionally include or cause an invoice to be generated for transmittal, as well as generate or cause a revenue sharing portion to be provided to the shop using system 10.

It should be understood that diagnostic computer tool 28 may not include all necessary calibration files 50c for all ADAS systems, such as for all makes, models and years of vehicles. In which case, checklist 26 will so indicate. Such unsupported ADAS systems may then require calibration via alternative means, such as directly through an OEM procedure that may be provided at an authorized dealer, or using specific equipment and systems, including calibration files, provided by a given OEM.

Still further, although diagnostic computer system 28 is shown and described in the illustrated embodiment above as including calibration files 50c retained within memory 50 of computer 30, it should be appreciated that some or all of the

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calibration function operations may be accomplished via a remote computer system 64. For example, calibration files 50c may be retained remotely and/or some aspects of the calibration function may be divided between a remote computer system 64 and the local computer 30. Still further, system list program 45 may communicate with remote server 64 to confirm that a calibration has been completed, such as for example if calibration files 50c are retained on server 64, including for example calibration programs. System 10 may additionally enable a mechanic 24 to obtain remote assistance, such as from a remote technician at computer system 64 via Internet connection 70.

FIG. 5 illustrates the general process of use of diagnostic system 10 for generating and using checklist 26. Step 100 is identification of ADAS systems present on a vehicle and step 102 is identification of damage areas on the vehicle, where the identification of the ADAS systems and damage areas may be performed as detailed above. The checklist 26 is then generated at step 104, such as via system list program 45. Subsequently, as illustrated at step 106, the checklist 26 is displayed to a mechanic, such as on a screen 66a of computer 30, along with displaying which of the ADAS systems listed on the checklist 26 are supported with calibration functions by the diagnostic computer tool 28. As understood from step 108, mechanic 24 may launch a calibration operation via checklist 26 and diagnostic computer tool 28. At step 110, upon successful completion of the calibration of an ADAS system listed on checklist 26, the status identification shown on checklist 26 may be toggled to complete. This step 110 may be done for each of the ADAS systems listed on checklist 26 as requiring calibration. Moreover, the identification of having completed calibration may be automatically generated by way of system list program 45 communicating with calibration files 50c, including if calibration files 50c are maintained on computer tool 28 and/or if maintained on remote server 64. And as illustrated at step 112, records of the completed calibrations may then be saved in a work file 83 regarding the repair work for the vehicle 22. Upon completion of the calibration operations, the mechanic 24 may then run a post scan using one of the diagnostic scanning application programs 53a, 53b for confirmation of completion of repairs to vehicle 22. Moreover, vehicle diagnostic computer tool 28 may additionally be configured to prevent performance of a post scan until confirmation has been entered that all required or suggested calibrations have been performed. As noted above, such confirmation of completion of a calibration may be by an operator toggling the indication on screen 66a, or may be automated via diagnostic computer tool 28 after successful calibration via program files 50c.

With further regard to determination of the ADAS systems present on vehicle 10 at step 100, system 10 may additionally employ or use a vehicle ADAS database 49, where database 49 includes information regarding all possible ADAS modules that may have been available for a given make, model and year of vehicle. For example, as noted above, diagnostic tool 28 may query electronic system 23 to obtain the VIN of vehicle 22, or the mechanic 24 may enter the VIN into diagnostic tool 28, such as a mechanic input 47. Alternatively, the mechanic may input the make, model and year of the vehicle 22 as a manual input. Based on the determined and decoded VIN and/or the make, model and year of the vehicle 22, database 49 is operable to determine what possible ADAS modules or systems may have been available for the vehicle 22.

In one embodiment, VIN database 37 and/or ADAS database 49 are located remotely from the vehicle repair

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facility and accessed by diagnostic tool 28, such as being retained within remote server 64 and accessed via a wireless Internet connection 70. Alternatively, databases 37 and 49 may be combined with database 50b, either remotely or locally. Likewise, system list program 45, diagnostic evaluation program 51, and/or diagnostic scan programs 53a, 53b may also be located remotely and be accessed by diagnostic computer tool 28. Alternatively, all or any of system list program 45, diagnostic evaluation program 51, and/or diagnostic scan programs 53a, 53b are contained within memory 50 of diagnostic tool 28, as discussed above. Still further, it should be additionally appreciated that programs 45, 50c 51, 53a, 53b may each comprise multiple subparts or routines that operate together. It should be further appreciated that inputs 21, such as vehicle safety system inputs 19, restraint deployment inputs 31 and/or damage inputs 35 may be variables that are used by or entered into system list program 45.

In a further particular alternative embodiment, system 10 may query the electronic system 23 of the vehicle 22 to obtain the part numbers of all ECUs resident on vehicle 22, where the ECU part numbers are readable via interface module 27 and the connection with OBD diagnostic port 32.

An exemplary embodiment of a database 50b configured as an ECU database is illustrated in FIG. 3B. As there shown, database 50b includes a listing of ECU part numbers 82a, 82b, 82c, such as for one particular make and model of vehicle, with the ECU part numbers being correlated with vehicle systems 81a, 81b, 81c, where the vehicle systems 81a, 81b, 81c in the illustrated embodiment are ADAS systems such as a forward facing lane keeping assist system, an adaptive cruise control system, and a collision avoidance system. It should be appreciated that although database 50b is illustrated in FIG. 3B to include three separate ECU part numbers associated with three separate vehicle systems, in practice a vehicle may be equipped with dozens or hundreds of ECUs where by the ECU database may include numerous listings of ECU part numbers as well as numerous vehicle systems. It should be appreciated that database 50b may include both ECU part numbers and addresses, which may be correlated or combined together.

The vehicle database 50b may further comprise a VIN/ECU part number database that includes a listing of ECU part numbers associated with a given VIN. Therefore, based upon a review of the VIN database and a review of the ECU part numbers associated with the current vehicle's VIN, a selection may be made between a third party supplied scanning program or a manufacturer supplied scanning program. For example, if the VIN database reveals that a given vehicle contains one or more ECU part numbers associated with ADAS ECUs, a decision can be made to select the more comprehensive manufacturer supplied scanning program. In the alternative, if a review of the VIN database reveals that the vehicle is not equipped with ADAS ECUs, the faster and less costly third-party supplied scanning program may be utilized. Such a decision may be aided if the VIN entry in the VIN database for an ADAS equipped vehicle includes at least one part number of an ADAS ECU associated with the VIN. Alternatively, the VIN entry in the VIN database for an ADAS equipped vehicle may include some other indication in the VIN database that the VIN is associated with an ADAS equipped vehicle. Therefore, if a vehicle's VIN entry in the VIN database does not include an ADAS ECU part number or some other indication that the VIN is for an ADAS equipped vehicle, the third-party supplied scanning program may be selected.

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Still further, system **10** may be configured to accept operator input to obtain vehicle data information. For example, system **10** may prompt an operator **24** to enter vehicle make and model information, such as via interface **66**. System **10** may then also provide step-by-step inquiries to operator **24** regarding systems installed on vehicle **22**, such as whether vehicle **22** includes particular cameras at particular locations, or specifically whether vehicle **22** has particular ADAS systems. Accordingly, system **10** instead of automatically launching a diagnostic scanning program **53a**, **53b** to be run after diagnostic evaluation program **51** queries vehicle electronic system **23** to determine which vehicle systems **81a**, **81b**, **81c** are present on vehicle **22**, or prior thereto, system **10** may instead provide a recommendation of whether to run an aftermarket diagnostic scan program **53a** or an OE diagnostic scan program **53b**, in particular based on various scan inputs **21**.

Although database **50b** is disclosed above as residing in diagnostic computer tool **28**, such a database may additionally or alternatively reside in the memory of a remote computer, such as a remote server **64**. If the database is located in a remote server **64**, the vehicle interface tool **28** may access the database via an Internet interface. Optionally, the database, such as a database found on the remote computer or server **64**, may be a compiled database of VINs and/or ECUs provided by a third party. Still further, the database found in the memory of the vehicle interface tool **28** may be a local database that comprises a portion of the information, where the database found on the remote computer or server **64** is remote and contains additional or alternative information. For example, the database may be progressively expanded to include those VINs and/or associated ECU part numbers for vehicles previously serviced at a particular service station or automotive repair business. Optionally, the database may be updated, such that an operator may add a new vehicle VIN and associated ECU part numbers. The database may thus be a tool for identifying which vehicles are equipped with ADAS subsystems.

Diagnostic tool **28** may, either in addition to the above noted various loaded diagnostic applications or in place thereof, be used to access remotely located diagnostic applications, such as that may reside on remotely located servers **64**. This may be done, for example, to avoid the need for obtaining and locally storing and maintaining diagnostic applications on diagnostic tool **28**. In the illustrated embodiment, diagnostic tool **28** includes an input/output (I/O) interface **66** for coupling to peripheral devices, such as one or more of a monitor, keyboard, mouse, and the like. In a further illustrated embodiment, the diagnostic tool **28** is implemented as a laptop computer with integrated monitor, keyboard, and mouse.

As an alternative to providing checklist **26** on screen **66** of computer **30**, diagnostic computer tool **28** may instead or in addition output the associated data of checklist **26** to a report. For example, an electronic data file may be generated by the vehicle diagnostic computer tool **28** that may be provided to another computer device and/or may be printed. The electronic data file may list the vehicle safety systems equipped on the vehicle requiring calibration. Still further, the output data file may also include calibration information for safety systems supported by the diagnostic computer tool **28**. For example, the calibration information may comprise calibration instructions that may be separately printed or displayed on another computer to provide human readable instructions for calibrating safety systems.

System **10**, in addition to performing scanning operations, may also be used for reprogramming of vehicle **22**, such as

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reprogramming selected ECUs, including reprogramming ECUs based on a determination of particular fault codes.

Although vehicle interface diagnostic tool **28** is discussed above as conforming with the SAE J2534 standard, it should be appreciated that alternatively configured vehicle diagnostic and programming tools may be employed within the scope of the present invention, including alternatively configured tools for alternative types of vehicles, such as alternative classes of vehicles. Accordingly, an interface tool may conform with the ISO 22900 standard, or RP1210 standard, or may operate under the ELM327 command protocol.

As noted, the software and/or hardware of diagnostic and programming tools may be required to be updated to operate with new vehicles and/or enable programming and diagnosing of existing vehicles. In the above noted embodiments the vehicle interface diagnostic tool **28** may be periodically updated via an Internet connection, or may be returned to the supplier for updating, including with regard to hardware updates. This may be done by the supplier of the vehicle interface diagnostic tool **28** whereby the local operator **24** need not spend time attempting to maintain the equipment.

In the illustrated embodiment vehicle interface diagnostic computer tool **28** is disclosed as including both the diagnostic evaluation program **51** as well as multiple diagnostic scanning programs **53a**, **53b**. It should be appreciated that alternative arrangements may be employed within the scope of the present invention. For example, program **51** and/or programs **53a**, **53b** may reside on different devices. For example, an operator **24** may first utilize one device with a diagnostic evaluation program **51** to determine which diagnostic evaluation program is appropriate for the vehicle **22**, and then select from alternative computer devices to run the determined diagnostic evaluation program. Still further, although diagnostic evaluation program **51** and diagnostic scanning programs **53a**, **53b** are illustrated and discussed herein as being separate programs, one or more of such programs may be combined together and operate as subroutines.

Changes and modifications in the specifically described embodiments can be carried out without departing from the principles of the present invention which is intended to be limited only by the scope of the appended claims, as interpreted according to the principles of patent law including the doctrine of equivalents.

The embodiments of the invention in which an exclusive property or privilege is claimed are defined as follows:

1. A method for identifying and presenting vehicle safety systems on a vehicle for calibration by an operator, said method comprising:

providing a vehicle diagnostic computer system comprising a vehicle diagnostic computer tool, wherein the vehicle diagnostic computer tool is configured to be connected to a diagnostic port of a vehicle to be in communication with an electronic system of the vehicle;

providing a system list program, wherein said system list program is operable to generate a list of vehicle safety systems equipped on the vehicle requiring calibration based on inputs to said system list program;

determining vehicle safety systems present on the vehicle via the vehicle diagnostic computer tool;

determining areas on the vehicle in which there is damage, wherein said areas comprise damage areas, and providing vehicle damage information inputs to the system list program based on the damage areas;

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generating a list of vehicle safety systems equipped on the vehicle requiring calibration via said system list program based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, wherein the list of vehicle safety systems equipped on the vehicle requiring calibration corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas; and providing to a user via the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration; wherein the vehicle diagnostic computer tool includes a display screen, and wherein the method further comprises displaying on the screen of the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration; wherein the list of vehicle safety systems requiring calibration comprise Advanced Driver Assistance Systems (ADAS), and wherein said vehicle diagnostic computer system further comprises electronic calibration data configured to provide calibration support for one or more of the listed vehicle safety systems requiring calibration; and wherein the list of vehicle safety systems requiring calibration additionally includes an indication of which of the vehicle safety systems requiring calibration are supported by said vehicle diagnostic computer system via the electronic calibration data, and wherein said displaying on the screen of the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration further comprises displaying an indication of which of the vehicle safety systems requiring calibration are supported by said vehicle diagnostic computer system, and wherein said method further comprises; accessing, by a user of the vehicle diagnostic computer tool, electronic calibration data for at least one vehicle safety system requiring calibration that is supported by said vehicle diagnostic computer system, wherein said accessing by the user comprises launching via the vehicle diagnostic computer tool electronic calibration data for the at least one vehicle safety system; and calibrating the at least one vehicle safety system based on the electronic calibration data, wherein said calibrating the at least one vehicle safety system comprises using by the user the vehicle diagnostic computer tool and electronic calibration data associated with the at least one vehicle safety system to calibrate the at least one vehicle safety system for repairing at least one of the damage areas of the vehicle.

2. The method of claim 1, wherein said vehicle diagnostic computer tool includes said system list program.

3. The method of claim 1, further comprising reading vehicle data information of the vehicle with the vehicle diagnostic computer tool, and wherein said determining vehicle safety systems present on the vehicle via the vehicle diagnostic computer tool comprises using the vehicle data information to determine the vehicle safety systems present on the vehicle.

4. The method of claim 3, wherein reading vehicle data information of the vehicle with the vehicle diagnostic computer tool comprises reading a vehicle identification number (VIN) of the vehicle.

5. The method of claim 4, wherein said determining vehicle safety systems present on the vehicle via the vehicle diagnostic computer tool comprises using a database to identify vehicle systems present on the vehicle.

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6. The method of claim 1, wherein the vehicle damage information inputs comprises at least one of; one or more digital images of the vehicle, and wherein image recognition software is used on the one or more digital images to determine damage to the vehicle; data from a collision estimating software; or data input by a user of the vehicle diagnostic tool.

7. The method of claim 1, wherein said displaying on the screen of the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration further comprises displaying on the screen of the vehicle diagnostic computer tool an indication of completion of calibration of vehicle safety systems upon completion of calibration thereof.

8. The method of claim 1, wherein the vehicle diagnostic computer tool is operatively connected with a remote computer system, and wherein the system list program interfaces with the remote computer system, with the remote computer system configured to confirm that all of the listed vehicle safety systems equipped on the vehicle requiring calibration have been calibrated.

9. The method of claim 1, further comprising running a vehicle diagnostic scan using said vehicle diagnostic computer tool, and wherein prior to running the vehicle diagnostic scan said vehicle diagnostic computer tool requires confirmation that all of the listed vehicle safety systems equipped on the vehicle requiring calibration have been calibrated.

10. The method of claim 9, wherein the vehicle diagnostic scan comprises a post scan after performing repairs on the vehicle.

11. A vehicle diagnostic system comprising:
a vehicle diagnostic computer tool, wherein the vehicle diagnostic computer tool includes a display screen, and wherein the vehicle diagnostic computer tool is configured to be connected to a diagnostic port of a vehicle to be in communication with an electronic system of the vehicle;
a system list program;
electronic calibration data configured to provide calibration support for one or more vehicle safety systems; wherein said vehicle diagnostic tool is configured to determine vehicle safety systems present on the vehicle, and wherein said vehicle diagnostic tool is configured to receive vehicle damage information inputs corresponding to areas on the vehicle in which there is damage, wherein said areas comprise damage areas;
wherein said system list program is configured to generate a list of vehicle safety systems equipped on the vehicle requiring calibration based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, wherein the list of vehicle safety systems equipped on the vehicle requiring calibration corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas;
wherein said vehicle diagnostic computer tool is configured to display on the screen of the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration and an indication of which of the vehicle safety systems requiring calibration are supported for calibration via the electronic calibration data; and
wherein said vehicle diagnostic computer tool is configured to enable a user to launch via the vehicle diagnostic computer tool electronic calibration data for each of the vehicle safety systems equipped on the vehicle

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requiring calibration and that are supported for calibration for repairing the damage areas of the supported vehicle safety systems.

12. The vehicle diagnostic system of claim 11, wherein the vehicle diagnostic computer tool comprises the system list program.

13. The vehicle diagnostic system of claim 11, wherein the vehicle diagnostic computer tool is configured to read vehicle data information of the vehicle to determine the vehicle safety systems present on the vehicle.

14. The vehicle diagnostic system of claim 13, wherein the vehicle data information of the vehicle comprises a vehicle identification number (VIN) of the vehicle, and wherein the vehicle diagnostic computer tool is configured to interface with a database to determine the vehicle safety systems present on the vehicle based on the VIN.

15. The vehicle diagnostic system of claim 11, wherein the vehicle damage information inputs comprise at least one of;

one or more digital images of the vehicle, and wherein image recognition software is used on the one or more digital images to determine damage to the vehicle; data from a collision estimating software; or data input by a user of the vehicle diagnostic tool.

16. The vehicle diagnostic system of claim 11, wherein the vehicle diagnostic computer tool is configured to allow a user to access the electronic calibration data with the vehicle diagnostic computer tool for calibration of vehicle safety systems requiring calibration that are supported by said vehicle diagnostic system.

17. The vehicle diagnostic system of claim 11, wherein said system further comprises diagnostic scan application programs, and wherein the vehicle diagnostic computer tool is operable to run a selected one of the diagnostic scan application programs to diagnose the vehicle, and wherein the vehicle diagnostic computer tool is configured to require confirmation that all of the listed vehicle safety systems equipped on the vehicle requiring calibration have been calibrated prior to the vehicle diagnostic computer tool running the selected one of the diagnostic scan application programs.

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18. The vehicle diagnostic system of claim 11, wherein said vehicle diagnostic computer tool is configured to display on the screen of the vehicle diagnostic computer tool an indicator display field associated with each listed vehicle safety system requiring calibration, wherein each indicator display field provides an indication whether the associated vehicle safety system has been calibrated.

19. The vehicle diagnostic system of claim 11, wherein said vehicle diagnostic computer tool is configured to display on the screen of the vehicle diagnostic computer tool a support display field associated with each vehicle safety system requiring calibration that is supported by said vehicle diagnostic computer system via the electronic calibration data, and wherein the support display field is selectively activatable by a user of the diagnostic computer tool, and wherein upon activation of a given support display field electronic calibration data comprising calibration instructions for the associated vehicle safety system are provided on the display screen of the diagnostic computer tool.

20. The method of claim 1, further comprising displaying on the display screen of the diagnostic computer tool an indicator display field associated with each listed vehicle safety system requiring calibration, wherein each indicator display field provides an indication whether the associated vehicle safety system has been calibrated.

21. The method of claim 1, further comprising displaying on the display screen of the diagnostic computer tool a support display field associated with each vehicle safety system requiring calibration that is supported by said vehicle diagnostic computer system via the electronic calibration data.

22. The method of claim 21, wherein the support display field is selectively activatable by a user of the diagnostic computer tool, and wherein upon activation of a given support display field electronic calibration data comprising calibration instructions for the associated vehicle safety system are provided on the display screen of the diagnostic computer tool.

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